

QUALITATIVE ANALYSIS WITHOUT INTEGRATING • DENNIS G. ZILL & DR. HAGHIGHAT LECTURE 2

Chapter 2.1: Direction Fields & Isoclines

Understanding the geometric compass of differential equations, constructing lineal elements, manual sketching using isoclines, and predicting trajectory behavior.

Zill §2.1 (pp. 65–70)

Lecture 2 Slides 9–13

Midterm 1 High-Yield

PLAIN-ENGLISH INTUITION

Even when an ODE is mathematically impossible to solve symbolically, the differential equation $\frac{dy}{dx} = f(x, y)$ gives the exact instantaneous tangent slope at every point (x, y) . By charting these slopes as tiny line segments (lineal elements), we construct a **direction field**. Following the slope vectors reveals the entire family of solution curves without doing any calculus!

FORMAL DEFINITIONS • SLOPE COMPASS

- **Lineal Element:** A short line segment centered at (x, y) with slope $m = f(x, y)$.
- **Direction Field:** A grid array of lineal elements showing the flow of the ODE.
- **Solution Trajectory:** Any valid integral curve $y = \phi(x)$ is everywhere tangent to the direction field segments.
- **Picard Uniqueness:** Where f and $\partial f / \partial y$ are continuous, solution curves can NEVER cross or touch.

THE METHOD OF ISOCINES

An **isocline** is a curve in the plane along which all tangent slopes share a constant value c :

$$f(x, y) = c \quad (c = \text{constant slope})$$

Manual Sketching Steps:

1. Choose simple constants: $c \in \{-2, -1, 0, 1, 2\}$.
2. Plot the curves $f(x, y) = c$.
3. Draw parallel hash marks with slope c all along each curve.
4. **Nullclines** ($c = 0$): Slopes are horizontal ($m = 0$).

Worked Example 1 • Hyperbolic Isoclines (Lecture 2 Slide 11)

CONCORDIA LECTURE

Problem: Sketch the direction field of $\frac{dy}{dx} = 0.2xy$ using isoclines.

1 Isocline Formula: $0.2xy = c \implies xy = 5c$.

- $c = 0$: $xy = 0 \implies$ The x -axis and y -axis are nullclines with horizontal slopes ($m = 0$).
- $c = 1$: $xy = 5 \implies y = 5/x$ (hyperbola in Q1 and Q3 with slope $m = +1$).
- $c = -1$: $xy = -5 \implies y = -5/x$ (hyperbola in Q2 and Q4 with slope $m = -1$).

2 Analytical Verification:

Separating variables: $\int \frac{dy}{y} = \int 0.2x \, dx \implies \ln |y| = 0.1x^2 + C_1 \implies y(x) = Ce^{0.1x^2}$.

The curves are bell-shaped exponentials centered at $x = 0$, exactly matching the upward and downward slopes indicated by the hyperbolic isoclines!

Worked Example 2 • Parallel Line Isoclines

EXAM CLASSIC

Problem: Determine the isoclines and sketch the direction field for $\frac{dy}{dx} = x - y$.

- Isoclines: $x - y = c \implies y = x - c$ (a family of parallel lines with slope $+1$).
- For $c = 0$, $y = x$ is a nullcline ($m = 0$). For $c = 1$, $y = x - 1$ has slope $m = 1$.
- **Remarkable Observation:** Along the line $y = x - 1$, the slope of the isocline is 1, and the lineal elements also have slope $m = 1$! Thus, the straight line $y = x - 1$ is an exact solution curve!

2 Autonomous Equations, Phase Lines & Asymptotic Stability

PLAIN-ENGLISH INTUITION

An equation is **autonomous** when rate of change depends solely on current state y : $\frac{dy}{dx} = f(y)$. Because x does not appear, slopes are strictly horizontal across space. We can collapse the entire 2D behavior into a single 1D **phase line** to classify whether equilibrium states attract or repel nearby trajectories over time!

CRITICAL POINTS & EQUILIBRIUM SOLUTIONS

- A number c is a **critical point** if $f(c) = 0$.
- The constant function $y(x) \equiv c$ is an **equilibrium solution**.
 - Horizontal lines $y = c$ divide the plane into invariant strips: a solution starting inside a strip can never cross an equilibrium boundary!

PHASE LINE CONSTRUCTION PROTOCOL

- Solve $f(y) = 0$ to find all equilibrium points.
- Plot critical points as dots on a vertical y -axis.
- Test a point y in each interval:
 - $f(y) > 0 \implies y' > 0 \implies$ draw arrow **UP** ($\uparrow$).
 - $f(y) < 0 \implies y' < 0 \implies$ draw arrow **DOWN** ($\downarrow$).

The Three Stability Classifications (Lecture 2 Slide 17 & Zill p. 69)

Stability Type	Phase Line Arrows	Behavior as $x \rightarrow \infty$	Physical Analogy
Attractor (Asymptotically Stable)	$\downarrow \bullet \uparrow$ (pointing toward c)	Nearby trajectories converge to $y = c$.	Marble at the bottom of a bowl; self-correcting system.
Repeller (Unstable)	$\uparrow \bullet \downarrow$ (pointing away from c)	Nearby trajectories diverge away from $y = c$.	Ball balanced on a sharp needle; any nudge causes escape.
Semi-Stable	$\uparrow \bullet \uparrow$ or $\downarrow \bullet \downarrow$ (same direction)	Attracts from one side, repels on the other.	One-way threshold; delicate inflection state.

Master Worked Problem · Multi-Point Stability Analysis

HIGH-YIELD EXAM

Problem: Find all critical points, sketch the 1D phase line, and classify the stability of $\frac{dy}{dx} = y^2(y - 1)(3 - y)$.

1 Find Critical Points:

$f(y) = y^2(y - 1)(3 - y) = 0 \implies y = 0, \quad y = 1, \quad y = 3.$

2 Sign Analysis:

- $y > 3$: Test $y = 4 \implies (16)(3)(-1) = -48 < 0 \implies \downarrow$
- $1 < y < 3$: Test $y = 2 \implies (4)(1)(1) = +4 > 0 \implies \uparrow$
- $0 < y < 1$: Test $y = 0.5 \implies (0.25)(-0.5)(2.5) = -0.31 < 0 \implies \downarrow$
- $y < 0$: Test $y = -1 \implies (1)(-2)(4) = -8 < 0 \implies \downarrow$

3 Classification:

- $y = 3$: Top $\downarrow$, Bottom $\uparrow$. Arrows point inward $\implies$ **Asymptotically Stable (Attractor)**.
- $y = 1$: Top $\uparrow$, Bottom $\downarrow$. Arrows point outward $\implies$ **Unstable (Repeller)**.
- $y = 0$: Top $\downarrow$, Bottom $\downarrow$. Arrows point down on both sides $\implies$ **Semi-Stable**.

Notice: The root $y = 0$ has multiplicity 2 (y^2), which never changes sign! By the Multiplicity Theorem, even powers are ALWAYS semi-stable!

THE DEADLY EXAM TRAP · EQUILIBRIUM CURVES VS ASYMPTOTES

Do NOT label $y = c$ as an asymptote! $y(x) \equiv c$ is a genuine, exact, horizontal solution curve. Other curves approach it asymptotically, but the line itself is a solution!